Block Two Synthesis: Reflux, from the Happy Spitter to the Refractory Adolescent.

Imagine a single clinic morning. A ten-week-old girl who spits after most feeds and cries in the evenings has been on omeprazole for three weeks, and her parents are not sure it helps. A seven-month-old boy has started vomiting for the first time and is slipping down his weight chart, and he has been referred as late reflux. A fifteen-year-old girl has heartburn four days a week despite high-dose omeprazole, and her endoscopy was normal. A sixteen-year-old boy whose oesophageal atresia was repaired at birth is about to move to adult care, and nobody has looked inside his oesophagus since he was five. And an eight-year-old girl with severe cerebral palsy is booked for a gastrostomy, and the surgeon asks whether to add an anti-reflux operation at the same time. Every one of these children carries the same word on the referral letter: reflux. And every one of them needs something different.

The main idea of this chapter is that reflux is a normal event in every infant and a disease in only a few. Almost every error we met in the thirteen lessons of this block came from treating the word instead of the phenotype: acid suppression for a crying baby, a proton pump inhibitor for a cough, an anti-reflux operation for a child whose real problem is the swallow. So the habit to build is simple. Before every prescription, ask which phenotype this is, what would prove it, and what would change if you were wrong.

This chapter brings the block together. It starts with a map of the phenotypes, then compares the three anchor documents that the plan for this block names: the 2018 joint guideline on paediatric gastro-oesophageal reflux from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, led by Rosen; the second version of the Lyon Consensus on the modern diagnosis of reflux disease, published in Gut in 2024 by Gyawali and colleagues; and the guideline from the National Institute for Health and Care Excellence in the United Kingdom, known as N G one, published in 2015 and updated in 2019. It then looks at the evidence as a whole, works through the five children from the opening, and closes with twenty pearls, the gaps in our knowledge, and what to do differently in clinic.

A few terms first. Gastro-oesophageal reflux, often shortened to G E R, means the passage of stomach contents into the oesophagus, which happens many times a day in everyone. Gastro-oesophageal reflux disease, G E R D, means reflux that causes troublesome symptoms or complications. A proton pump inhibitor, or P P I, blocks the acid pump in the stomach lining, and a histamine-two receptor antagonist is an older, weaker acid blocker. Multichannel intraluminal impedance combined with p H monitoring, usually called p H impedance, is a catheter study that detects every reflux episode, acid or not, over twenty-four hours. The acid exposure time is the percentage of that time during which the oesophagus is acidic. And the Los Angeles classification grades erosive oesophagitis at endoscopy from A, the mildest, to D.

Part one. Every phenotype has its own test and its own treatment

The first phenotype is physiological reflux in the infant. The baby regurgitates, is thriving, and has no warning signs. The best test is the history and the growth chart, and the treatment is explanation, attention to feed volume, and thickening if needed. What does not belong here is any acid-suppressing drug.

The second is infant reflux disease, meaning troublesome symptoms or complications. Here the tools are thickened feeds and a two- to four-week trial of an extensively hydrolysed formula, because cow's milk protein allergy can look exactly like reflux. A proton pump inhibitor belongs only to proven oesophagitis, and an empirical course must never be used as a diagnostic test in an infant.

The third is erosive reflux disease in an older child, defined by erosive oesophagitis of Los Angeles grade B or worse, a peptic stricture or Barrett's oesophagus. It is proved by endoscopy with biopsies and treated with a four- to eight-week course of a proton pump inhibitor, then weaned, rather than continued without review.

The fourth is non-erosive reflux disease, in which the mucosa looks normal but the acid exposure time is abnormal on p H impedance done off treatment. It responds to acid suppression, and surgery is reserved for disease that is proven and refractory.

The fifth and sixth phenotypes are the ones in which acid is not the problem. In reflux hypersensitivity the acid exposure is normal, but the child's symptoms line up with normal reflux episodes, which we measure with the symptom index and the symptom association probability. In functional heartburn the acid exposure is normal and symptoms do not line up with reflux at all. Both are disorders of oesophageal sensation. They are treated with gut-brain therapies and sometimes with neuromodulators, not with higher doses of acid suppression and never with fundoplication.

The seventh is extra-oesophageal reflux: cough, asthma or hoarseness without typical reflux symptoms. In children these are rarely caused by reflux. The right tests are a swallow study and an airway work-up, and the wrong move is an empirical proton pump inhibitor.

Part two. The three anchor documents

The 2018 North American and European guideline covers infants to adolescents but explicitly excludes children with neurological impairment, repaired oesophageal atresia and cystic fibrosis, who have their own documents. It used a formal grading process and was honest that most of the evidence is of low quality. The line to remember from it is this: do not use histamine-two blockers or proton pump inhibitors for crying, distress or visible regurgitation in otherwise healthy infants, and do not use an empirical proton pump inhibitor as a diagnostic test in infants.

The Lyon Consensus version two is an adult document about how to read endoscopy and reflux physiology. Its key numbers are these: off treatment, an acid exposure time above six percent is conclusive for reflux disease, below four percent is physiological, and between four and six percent is inconclusive. More than eighty reflux episodes a day supports the diagnosis and fewer than forty argues against it. Los Angeles grade B oesophagitis is now conclusive. The important caveat is that Lyon two never mentions children, so every paediatric use of its thresholds is an extrapolation.

The NICE guideline covers infants, children and young people in the United Kingdom and was built on systematic reviews with health-economic modelling. It says not to offer acid suppression for overt regurgitation on its own, but it allows a four-week trial of a proton pump inhibitor or histamine-two blocker in an infant with regurgitation together with unexplained feeding difficulty, distressed behaviour or faltering growth, followed by review. Two supporting documents complete the picture: the 2026 Rome five paediatric criteria, which for the first time give children their own definitions of reflux hypersensitivity and of reflux-negative oesophageal pain, and the 2019 chronic cough guideline from the American College of Chest Physicians, which says not to use acid suppression solely for chronic cough in children.

Part three. Where the documents disagree

For infants, the North American and European guideline and NICE agree on the healthy spitter, who needs reassurance and no tests or drugs. They agree that acid suppression should not be used for regurgitation alone. They agree on sleep: supine, on a flat surface, with no positioning therapy. But they part on three points. The material difference is the distressed infant: the joint guideline recommends against acid suppression, while NICE allows a four-week trial in selected infants. They differ on the milk trial: the joint guideline uses a two- to four-week trial of an extensively hydrolysed formula or maternal elimination before any drug, while NICE has no hydrolysate step for reflux. They differ on alginates: against chronic use in the joint guideline, a one- to two-week trial in NICE. There is a material gap on thickeners in preterm infants, where the joint guideline states limits and NICE says nothing, despite cases of necrotising enterocolitis and deaths that led the United States Food and Drug Administration to warn against a xanthan gum thickener in 2011. And neither document gives a number for overfeeding. In short, before any drug, North America and Europe test for milk allergy, and the United Kingdom tries an alginate.

After infancy the comparison is between the paediatric guideline and adult numbers. On an empirical proton pump inhibitor trial for typical heartburn in an older child, the documents align: four to eight weeks in the joint guideline, four weeks then review in NICE, and Lyon adds that response alone is not conclusive. On abnormal acid exposure they differ: the paediatric guideline still uses the older reflux index, above ten percent under one year and above five percent over one year, while Lyon uses its adult thresholds of six and four percent. On airway symptoms alone they align: no proton pump inhibitor as a test, and test before

treating. On impedance metrics such as the number of reflux episodes and the mean nocturnal baseline impedance, there is a gap, because no paediatric normal values exist. On functional phenotypes the difference is material, because the 2018 guideline borrowed adult labels and the 2026 Rome five paediatric criteria now define them for children. On surgery they align: operate only for proven refractory disease, and in neurologically impaired children do not routinely add a fundoplication to a gastrostomy. The material gap after infancy is that adolescents are being tested against adult numbers that were never validated in them.

Part four. The block in one chart, and four numbers

Across the thirteen lessons of this block, the guideline comparison tables asked eighty-nine questions. Thirty-seven, or forty-two percent, received the same answer from the documents compared. Twenty-six, or twenty-nine percent, differed in detail. Fourteen, or sixteen percent, had no data anywhere, a true gap. And twelve, or thirteen percent, showed a material difference that would change care. For comparison, in the first block on the diagnostic toolkit, forty-six percent of questions were aligned. So agreement is less than half the story. The documents agree well on what not to do: no acid suppression for spitting, no positioning, no proton pump inhibitor for cough. They diverge on what to do next, and they fall silent on paediatric normal values, on how to stop a proton pump inhibitor, and on the high-risk groups. Most of the material differences concern a single question: when, if ever, to suppress acid in an infant.

Four numbers from the block change what happens next to a real child. The first is fifty-four percent equals fifty-four percent. In the large placebo-controlled trial of lansoprazole in infants by Orenstein and colleagues in 2009, fifty-four percent responded on the drug and fifty-four percent on placebo, while serious adverse events, mostly chest infections, occurred in ten infants on the drug and two on placebo. The second is the Lyon pair of six percent and four percent for acid exposure, which means a normal acid exposure time is a diagnosis, not a failed test. The third is twelve point four percent, the proportion of paediatric endoscopies that show erosive oesophagitis in a large American study by Gilger, together with confirmed Barrett's oesophagus in only one of eight hundred and forty otherwise healthy children. The fourth is eighty-three point nine percent, the proportion of coughs in coughing children that were independent of reflux on p H impedance. Behind each number, know the denominator: paediatric p H norms came from children who already had symptoms, endoscopy rates come from referred children, and cough data come from children already labelled with reflux.

Early acid suppression is not free. Large observational studies have found that a proton pump inhibitor in the first six months of life was associated with an adjusted hazard ratio for food allergy of about two point six. Proton pump inhibitor use in childhood was associated with asthma at a hazard ratio of about one point six. In one point two six million French children, proton pump inhibitor use was associated with serious infections at a hazard ratio of one point three four. And use in the first year was associated with fractures at a hazard ratio of about one point two. In very-low-birth-weight infants given ranitidine, infection occurred in thirty-seven

percent versus ten percent without it, necrotising enterocolitis was more than six times as likely, and death occurred in nine point nine percent versus one point six percent. A Cochrane review of proton pump inhibitors for cough found one extra adverse event for every eleven children treated for four weeks. None of these observational signals proves harm, and confounding by indication is possible. But each one is weighed against a benefit that, in infants, the trials did not show.

Part five. Five difficult cases

The first child is the ten-week-old girl on omeprazole. She weighs four point eight kilograms, is on the fiftieth centile and rising, spits after most feeds and cries in the evenings. Her parents give her nine hundred millilitres of formula a day, which is about one hundred and ninety millilitres per kilogram, well above the usual one hundred and fifty. Regurgitation peaks at sixty-seven percent of infants at four months and resolves in about ninety percent by one year. Lansoprazole did no better than placebo, omeprazole lowers acid without changing crying, and early acid suppression carries the allergy and infection signals we have just discussed. Both guidelines say no acid suppression for regurgitation alone. The decision is to stop omeprazole, bring the volume back toward one hundred and fifty millilitres per kilogram per day in smaller feeds, thicken the feeds, keep a two- to four-week hydrolysed formula trial in reserve if symptoms persist, and reinforce supine sleep.

The second child is the seven-month-old boy whose vomiting began three weeks ago. He never regurgitated before, his weight has fallen from the fiftieth to the twenty-fifth centile, he is more irritable, and he had one day of fever. The vomit is not green and the fontanelle is soft. The key fact is that vomiting that starts after six months of age, or persists beyond twelve to eighteen months, is not physiological reflux. NICE advises testing the urine in late-onset regurgitation, faltering growth or distress. Green vomit would be surgical and would need a same-day contrast study. So the decision is a urine culture, electrolytes and kidney function, a measured head circumference and a neurological examination, a contrast study if any vomit is bilious, and no proton pump inhibitor.

The third child is the fifteen-year-old girl with heartburn and a normal endoscopy. Biopsies showed no eosinophilic oesophagitis. Off treatment, her p H impedance showed an acid exposure time of two point eight percent and thirty-eight reflux episodes, but her symptom index was seventy-five percent and the symptom association probability ninety-eight percent. By Lyon criteria an acid exposure below four percent is physiological, and fewer than forty episodes argue against reflux disease. Normal acid with a positive symptom association means reflux hypersensitivity, now defined for children by Rome five. After a normal endoscopy in children with heartburn, functional heartburn accounts for thirty-eight to forty-four percent and hypersensitivity for twenty-one to twenty-nine percent. In adults with hypersensitivity, citalopram reduced persistent symptoms from about sixty-seven percent to thirty-eight and a half percent; there is no paediatric trial. The decision is to taper the proton pump inhibitor, explain oesophageal sensitivity, offer gut-brain therapy and consider a neuromodulator, and

avoid fundoplication.

The fourth child is the sixteen-year-old with repaired oesophageal atresia, three dilations in infancy, occasional food sticking, no heartburn, no medication, and no endoscopy since age five. Symptoms do not predict Barrett's oesophagus in these patients; in one series of adolescents with atresia, forty-three percent had Barrett's, mostly of the gastric type. The 2016 joint guideline on oesophageal atresia recommends endoscopy off acid suppression, before the age of ten, and at transition to adult care. In adults after atresia repair, Barrett's has been reported in about six point six percent and squamous cancer in about zero point seven percent. And any dysphagia or stricture deserves biopsies for eosinophils. The decision is endoscopy with biopsies now, a check for eosinophilic oesophagitis, and a transition letter that names adult surveillance every five to ten years.

The fifth child is the eight-year-old with cerebral palsy at the most severe functional level, two pneumonias this year, coughing and gurgling with oral feeds, and a gastrostomy booked. She retches when constipated, and she has had no p H impedance or endoscopy. The 2017 European guideline on neurologically impaired children recommends, strongly, against a routine fundoplication at the time of gastrostomy. Without a wrap, only about one child in ten needed one later. Between fundoplication and gastrojejunal feeding there are no randomised trials, and the 2026 Cochrane review found no clear difference in pneumonia or death. Aspiration from above, from the mouth and pharynx, needs a swallow study, not a wrap, and retching predicts that a wrap will fail. The decision is a gastrostomy alone, a videofluoroscopic swallow study, p H impedance on feeds if pneumonias continue, and gastrojejunal feeding or a fundoplication only for proven refractory reflux disease.

Part six. Twenty pearls

Here are the twenty pearls of the block. One: daily regurgitation affects about half of infants at up to three months, sixty-seven percent at four months, and five percent at ten to twelve months. Two: transient relaxations of the lower oesophageal sphincter carry eighty-two to ninety-four percent of infant reflux, and infants with reflux disease have more acid in each event, not more relaxations. Three: reflux disease means troublesome symptoms or complications, and regurgitation alone is neither necessary nor sufficient. Four: colic affects seventeen to twenty-five percent of infants in the first six weeks, and it is not reflux. Five: green vomit is surgical, and volvulus is present in about eighty percent of infants with malrotation who vomit bile. Six: thickening reduces regurgitation by about two episodes a day, with a number needed to treat of five, but does not change the p H reflux index. Seven: no xanthan gum thickener in preterm infants. Eight: the milk trial is two to four weeks of extensively hydrolysed formula or maternal elimination, followed by a re-challenge. Nine: sleep supine on a flat surface; raising the head of the cot does not work, and inclined sleepers have killed infants. Ten: lansoprazole fifty-four percent, placebo fifty-four percent, serious adverse events ten versus two.

Eleven: by Lyon two, off treatment, an acid exposure above six percent is conclusive and below four percent is physiological, and Los Angeles grade B is now conclusive. Twelve: a response to a proton pump inhibitor is weak evidence, with a sensitivity of about seventy-eight percent and a specificity of about fifty-four percent. Thirteen: erosive oesophagitis is seen in twelve point four percent of paediatric endoscopies, and Barrett's in one of eight hundred and forty healthy children. Fourteen: give a proton pump inhibitor thirty to sixty minutes before breakfast, at about one milligram per kilogram per day, and remember that children who are ultra-rapid metabolisers through the enzyme C Y P two C nineteen may need about double the dose. Fifteen: after eight weeks of a proton pump inhibitor, forty-four percent of healthy volunteers had reflux symptoms on stopping, compared with fifteen percent after placebo, which is rebound. Sixteen: in a lower-risk brief resolved unexplained event, do no reflux tests and give no acid suppression; fewer than three percent of cardiorespiratory events in preterm infants followed reflux. Seventeen: after a normal endoscopy, functional heartburn accounts for thirty-eight to forty-four percent of children with heartburn, hypersensitivity for twenty-one to twenty-nine percent, and non-erosive disease for about twenty-seven percent. Eighteen: eighty-three point nine percent of coughs are independent of reflux, and the chest physicians say no acid suppression for cough alone. Nineteen: no routine fundoplication at gastrostomy in neurologically impaired children, and in a randomised trial the Nissen wrap failed in five point nine percent compared with fifteen point nine percent for the partial Thal wrap. Twenty: a prophylactic proton pump inhibitor after atresia repair did not prevent strictures, forty-four point eight versus forty-four point one percent.

Part seven. What we still do not know

The block also exposed what is missing. There are no paediatric normal values for p H impedance, so every study is read by judgement or with adult numbers; healthy volunteer cohorts by age band would close that gap. There is no threshold for overfeeding in infants, so it is diagnosed by feel. Nobody has shown how best to stop a proton pump inhibitor in a child, and tapering, rebound warnings and rescue plans are all borrowed from adults. Neuromodulators for reflux hypersensitivity and functional heartburn have been tested only in adults. Fundoplication and gastrojejunal feeding in neurologically impaired children have never been compared in a randomised trial. The 2016 advice to give a year of prophylactic acid suppression after atresia repair has been contradicted by pooled data and is awaiting an update. The harms of early acid suppression rest on observational signals. And there is no agreed schedule for adult surveillance after atresia repair, which makes the transition letter more important than ever.

What to do differently in clinic

Here are eight habits to take from the block. Name the phenotype in the letter before naming a drug. Plot the growth chart and calculate millilitres per kilogram per day for every spitting infant. Never start a proton pump inhibitor in an infant for crying or spitting, and stop the ones

you inherit, with a plan. Use a single two- to four-week hydrolysed formula trial with a planned re-challenge, rather than repeated formula switches. Test before escalating, with p H impedance off treatment and endoscopy with biopsies for eosinophils. Read p H impedance with the Lyon numbers, but say aloud that they are adult numbers. Send the coughing infant for a swallow study before any reflux test. And write a stop date and a review date on every acid suppression prescription. If you do these eight things, most of the reflux in your clinic will turn out to be something else, and the children who truly have reflux disease will be found and treated properly.